Atoms and the Period Square

Learning Objective

To understand how atoms are represented on the periodic table.

Success Criteria: I can...

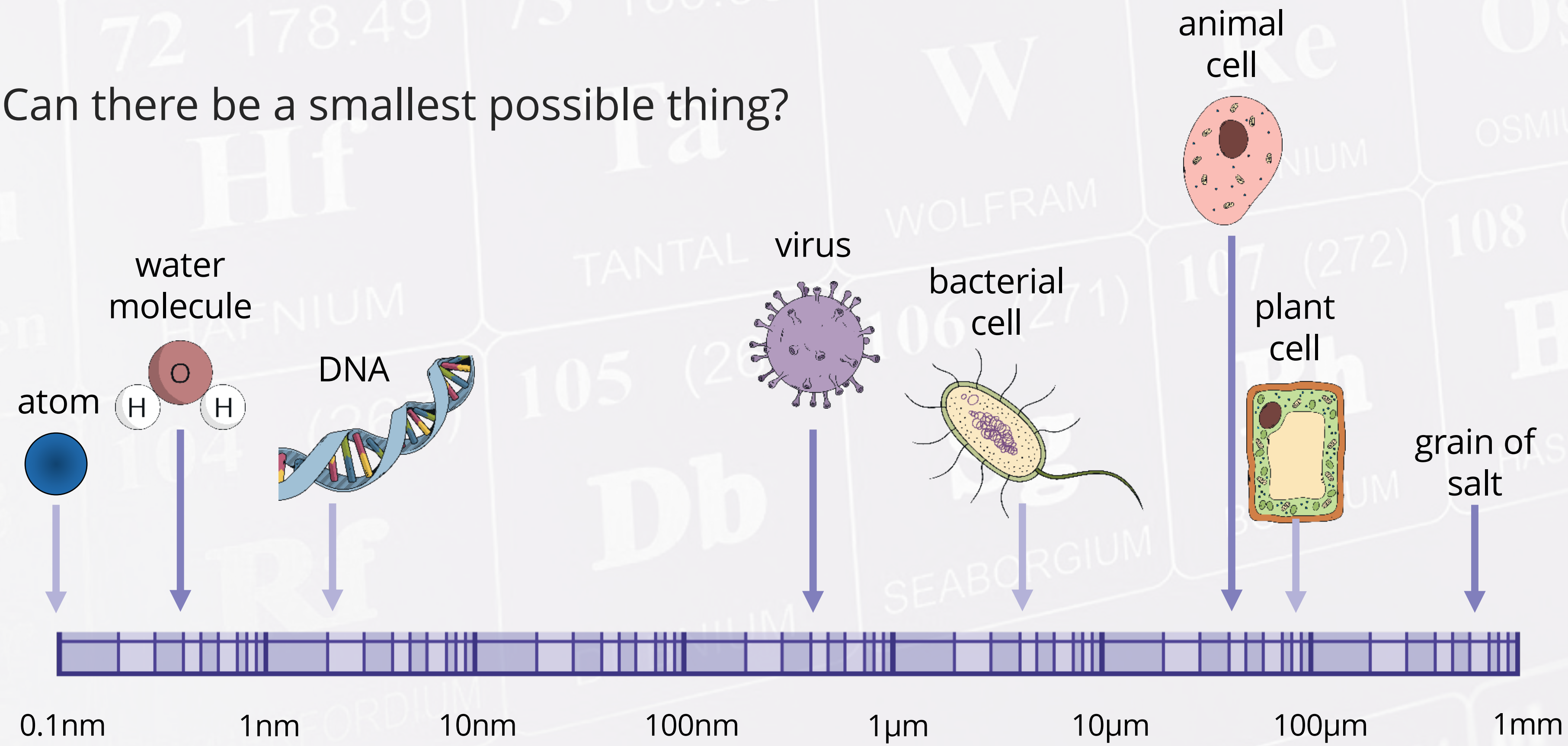
- describe atoms and elements
- describe the Bohr atomic model
- identify elements on the periodic table
- Draw a Bohr model for the first 20 elements

What Do You Think?

What is the smallest thing you can see?

What is the smallest thing you know about?

Can there be a smallest possible thing?



Elements

An element is a substance that cannot be broken down into other substances. They are made of only one kind of atom.

There are 94 naturally occurring elements like gold and oxygen.

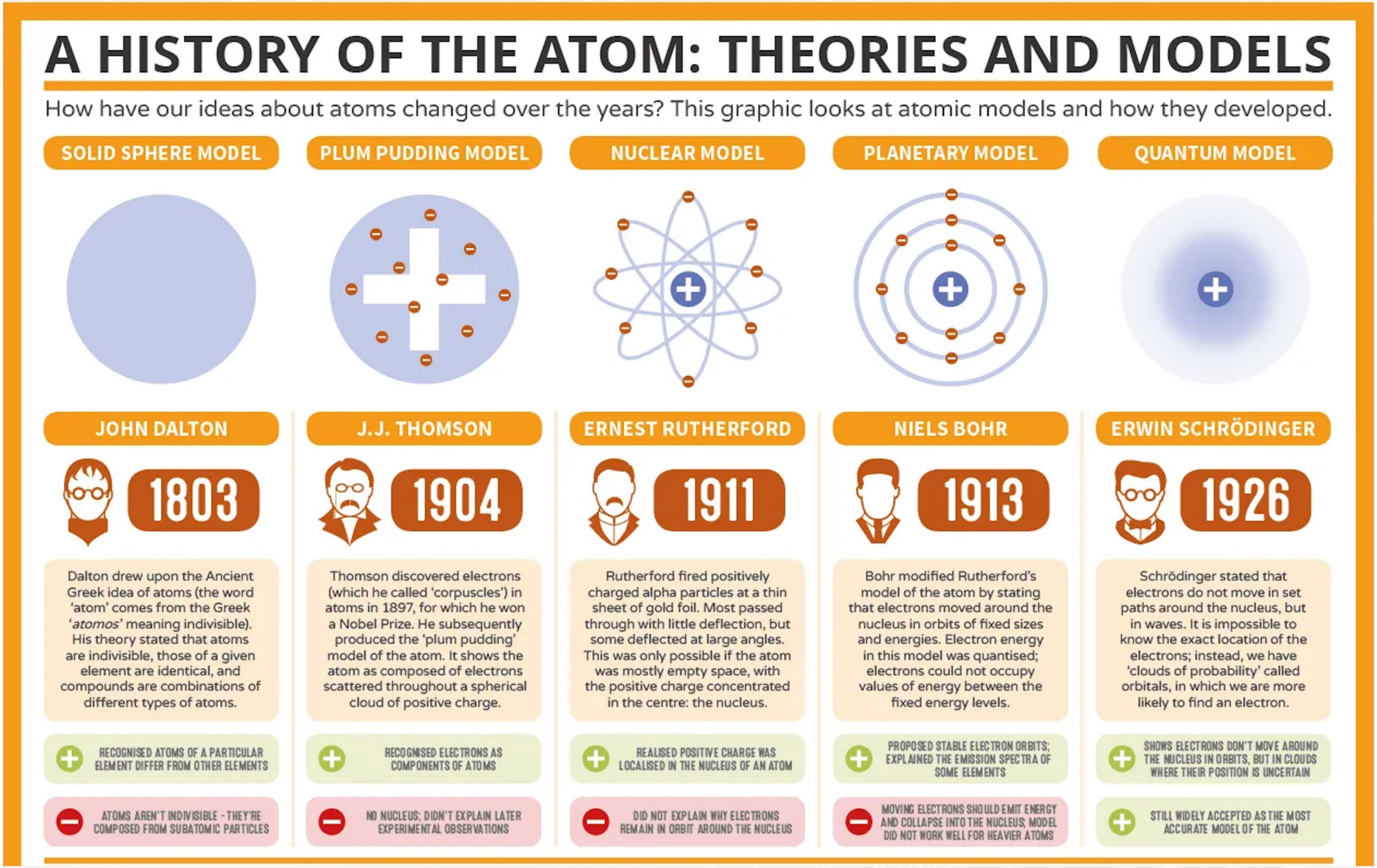
Some of the heavier elements have been created by people in a nuclear reactor or particle accelerator

Gold is a naturally-occurring element made in exploding stars

CERN's Large Hadron Collider (LHC) is the most powerful particle accelerator humans have built so far. It is buried underground near Geneva, Switzerland.

Element: A substance made of only one type of **atom**.

How atomic models have changed over time



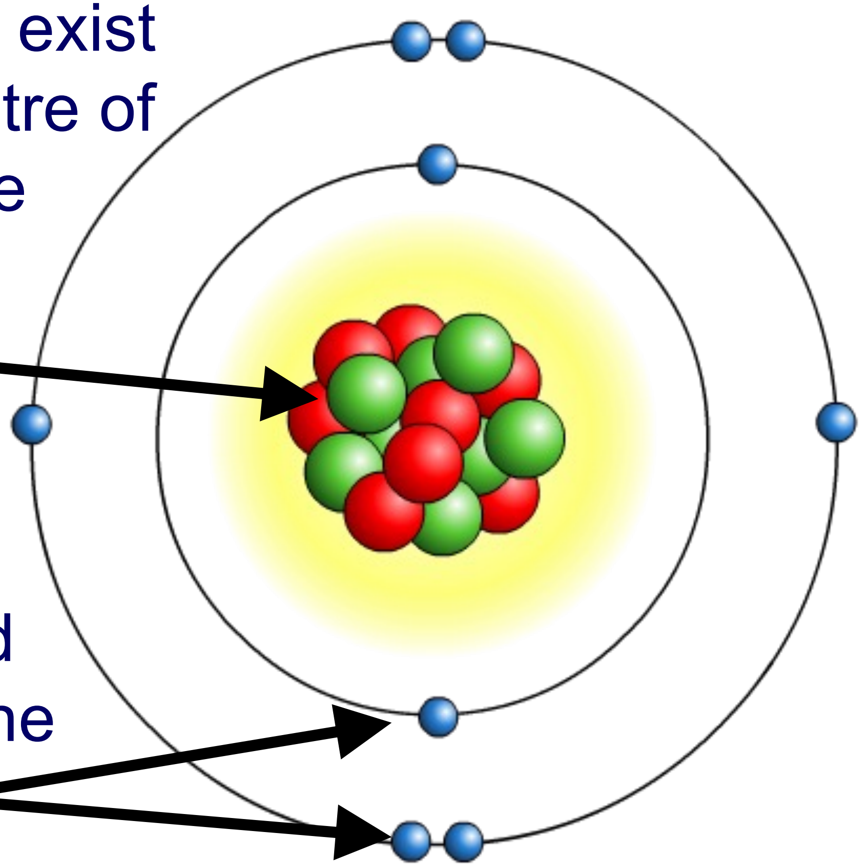
There are 3 subatomic particles that make up atoms. Together, they let the atom <u>have mass</u> and <u>take up space</u>

proton

neutron

electron

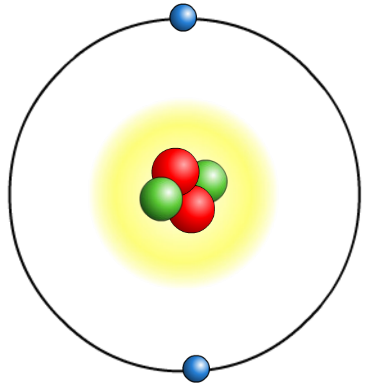
The protons and neutrons exist in a dense core at the centre of the atom. This is called the **nucleus**.



The electrons are spread out around the edge of the atom. They orbit the nucleus in layers called **shells** or orbitals.

Two important properties of protons, neutrons and electrons that are especially important:

- mass
- electrical charge.



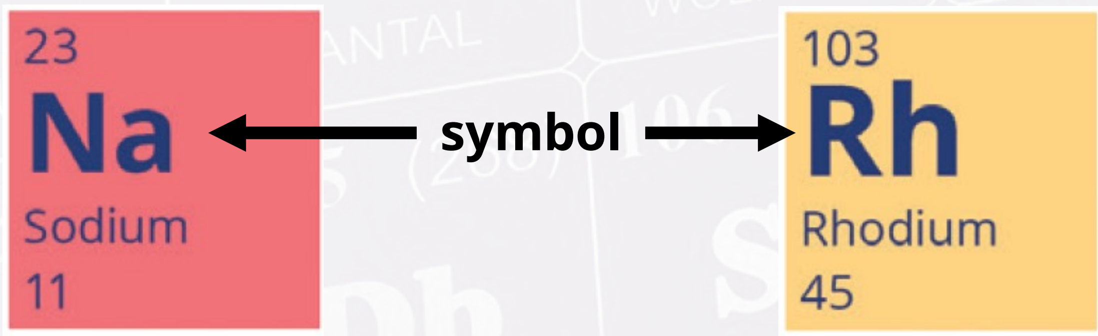
Particle	Mass	Charge
proton	1	+1
neutron	1	0
electron	almost 0	-1

The atoms of an element contain equal numbers of protons and electrons and so have no overall charge.

Chemical Symbols

Each element is represented by a symbol.

The symbol comes from the first letter or letters of its name. For elements discovered early on, the symbol usually comes from its Latin or Greek name. For example the symbol for sodium is Na, which comes from the Latin 'natrium'.



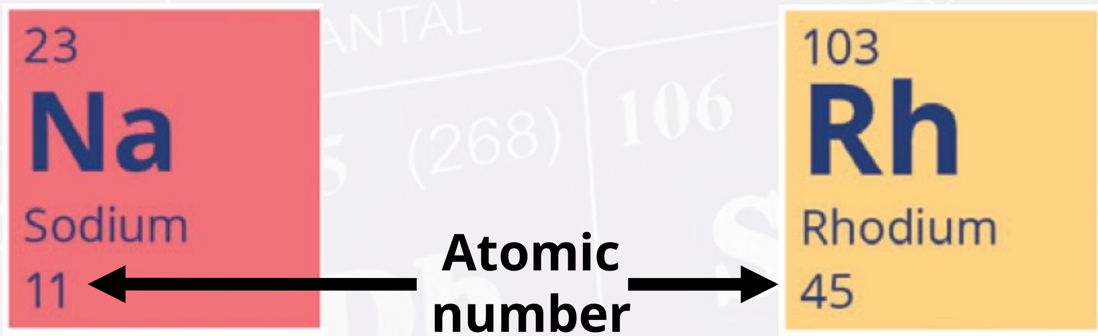
The first letter of the symbol is always capitalised. Any following letters are lower case. The symbol for each element can be found on the periodic table.

Why is it useful to use a symbol rather than just writing the name?

Atomic Numbers

The atomic number tells us the number of protons in an element.

An atom in nature may gain or lose electrons. That is called an **ion**.
An atom may also gain or lose neutrons, which is called an **isotope**.
What would happen if an atom gained or lost a <u>proton</u>?



It would be a totally different element with different chemical properties!

Can atoms of the same element have different numbers of electrons?

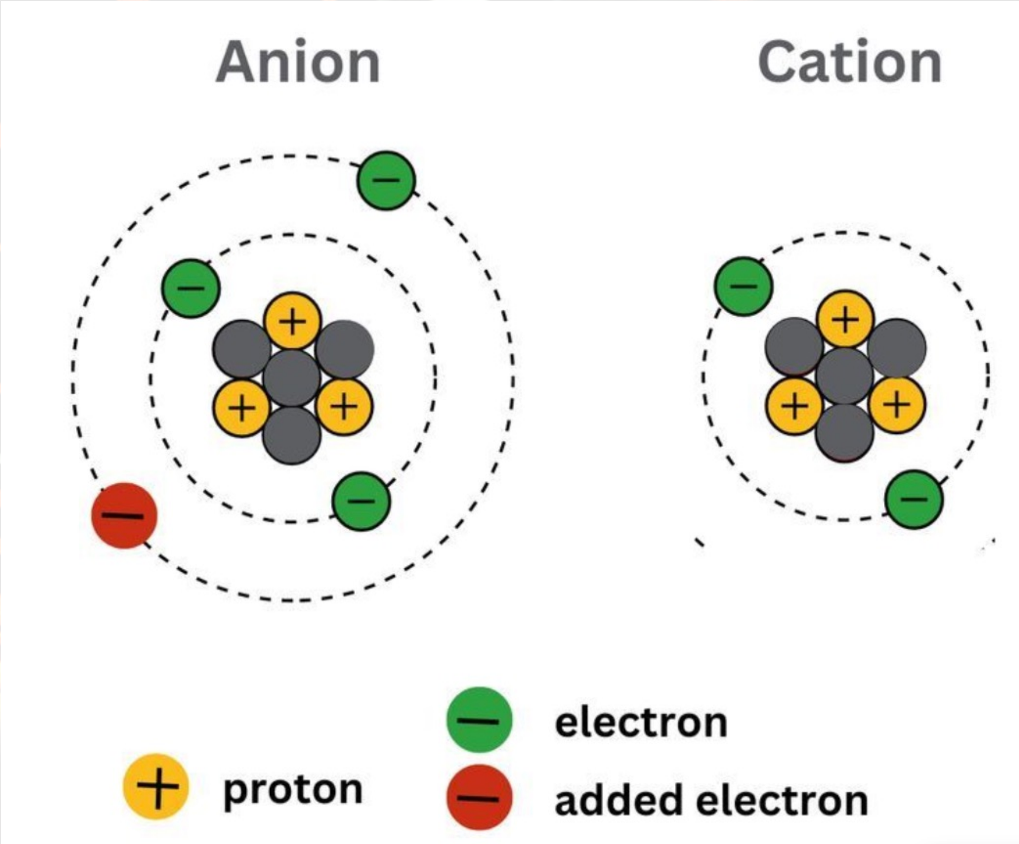
Atoms may gain or lose electrons. This is called an **ion**.

Ions are not electrically balanced.

An atom that gains electrons will have a negative charge. This is called an **anion.**

An atom that loses electrons will have a positive charge. This is called a **cation**.

3 Protons
-4 Electrons
-1 charge



3 Protons
-2 Electrons
+1 charge

BEYOND

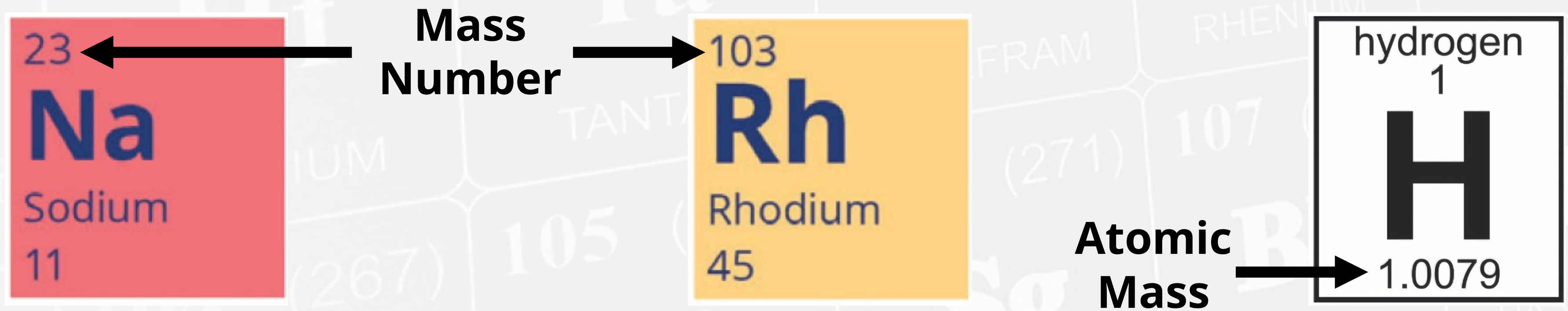
Atomic Mass & Mass Number

Atomic mass, sometimes called atomic weight, is the **average** mass of an element's atoms found in nature. Elements of the same atom can have different masses depending on the number of neutrons there are.

Atoms with **different numbers of neutrons** are called **isotopes**.

Generally, the electrons are too small to measurably affect the atomic mass.

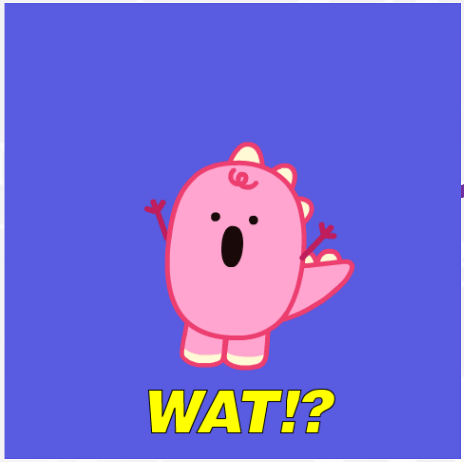
The **mass number** is a count of the total number of protons and neutrons in an atom's nucleus. It is **always a whole number**.

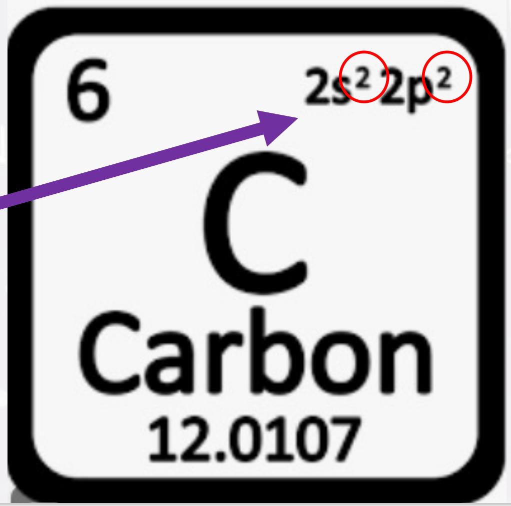


Example: Hydrogen has three natural isotopes: 1H, 2H, and 3H. Each isotope has a different mass number.

1H has 1 proton; its mass number is 1. 2H has 1 proton and 1 neutron; its mass number is 2. 3H has 1 proton and 2 neutrons; its mass number is 3. 99.98% of all hydrogen is 1H.1 It is combined with 2H and 3H to form the **average value of atomic mass of hydrogen, which is 1.00784 g/mol.**

But wait, there's more!

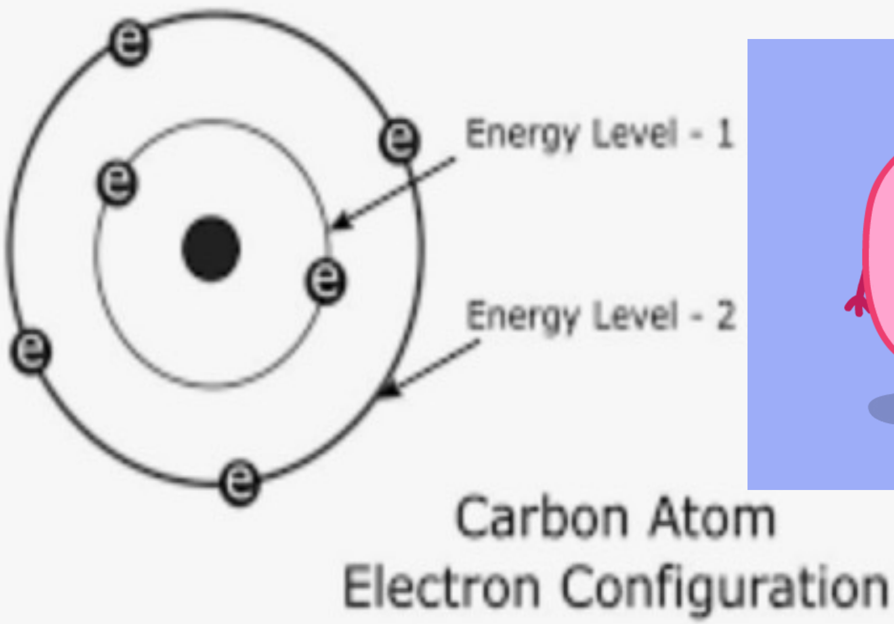




This is the **electron configuration.**

It tells us where the electrons can be found. Our Bohr mode shows the electron as a circle on an orbit. Each **orbital shell**, or energy level, can only hold a certain number of electrons.

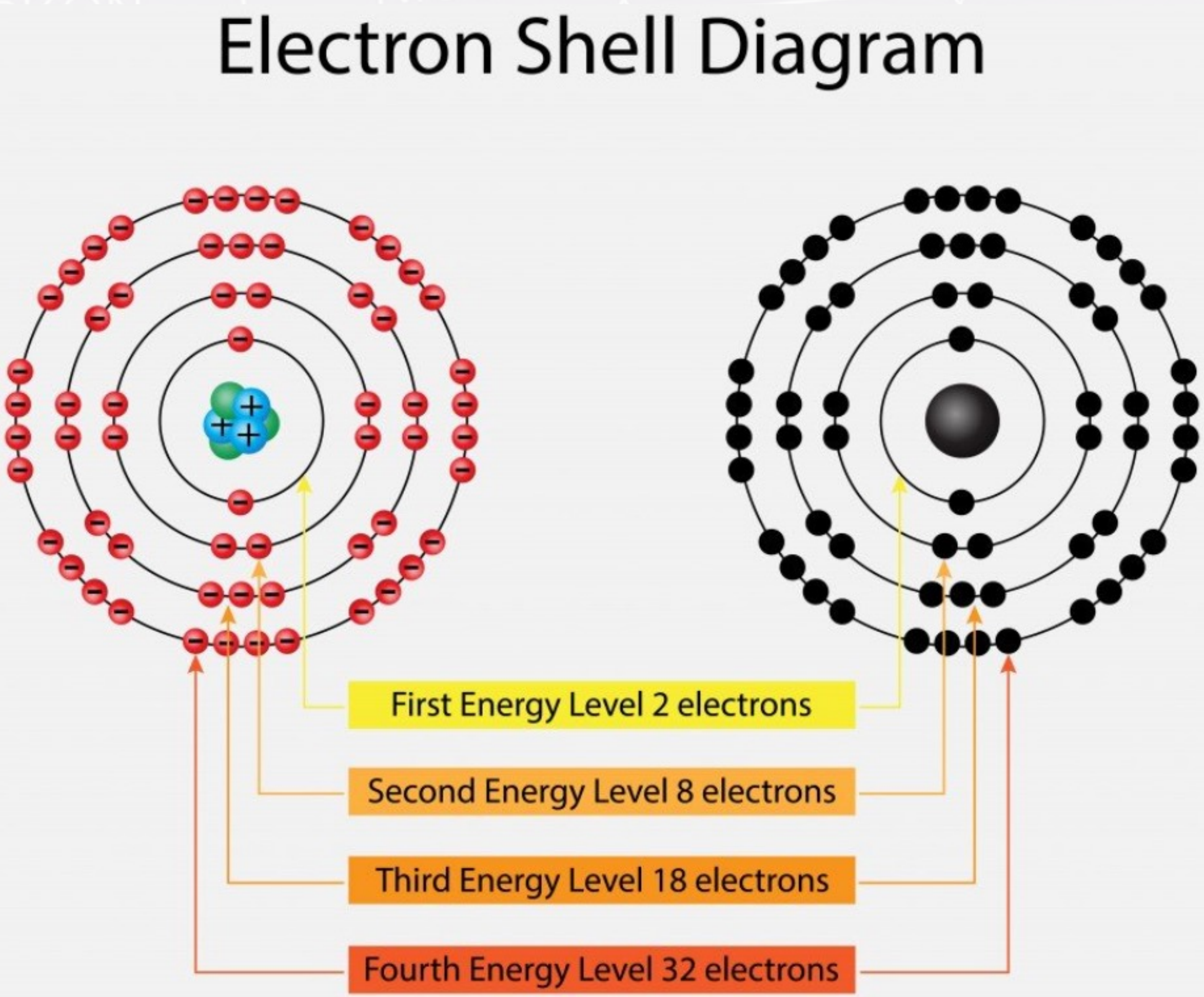
Use the super script numbers to count the outer, **valence** electrons. In balanced atoms, the electrons will equal the number of protons!



Don't worry about the S and P orbitals for now. The important thing to understand is that each orbital only holds a set number of electrons.

How many electrons fit in each orbital shell?

Each orbital shell, or energy level, holds a maximum number of electrons between 2 and 32. The electrons in the outer-most shell are called **valence** electrons.



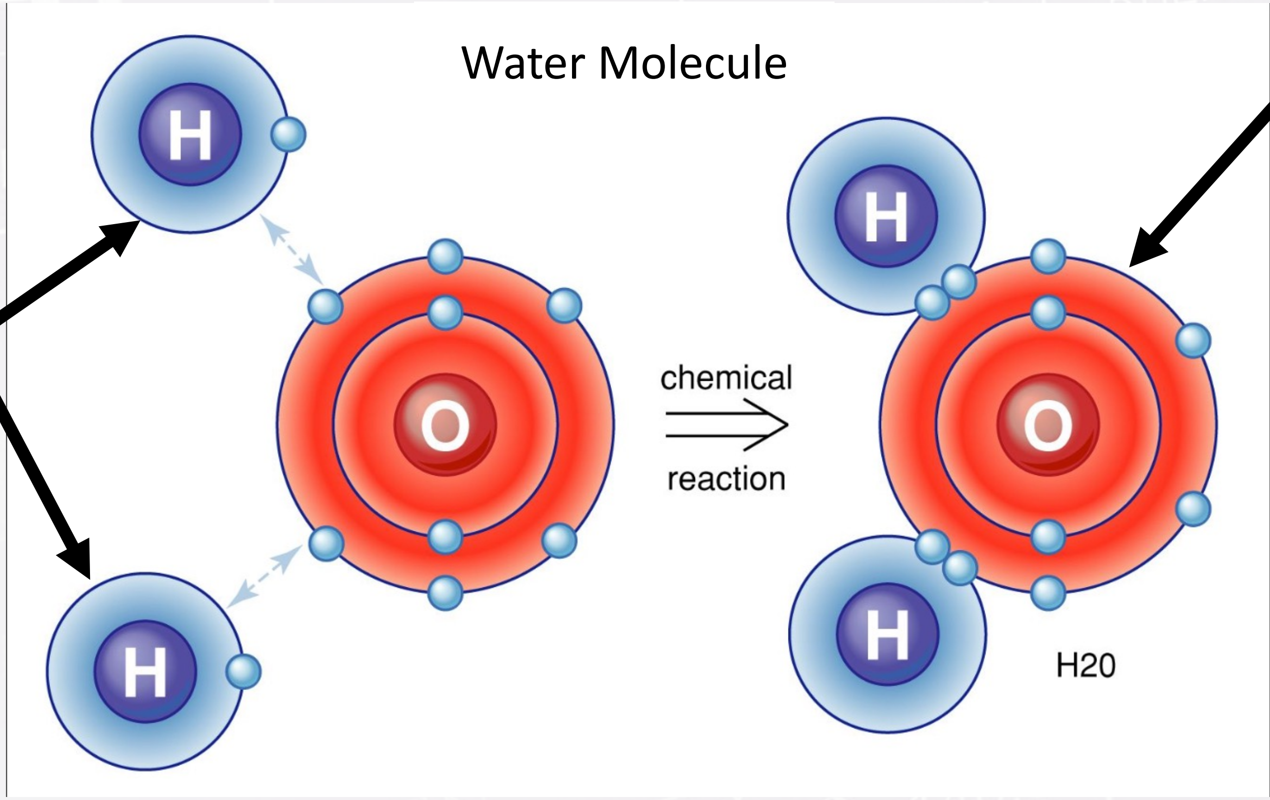
Bohr Model and Bonds

We can use the energy levels of the electrons to tell us how an atom likes to bond.

Atoms like to fill up their outer shell, so if an atom has 1 or 2 electrons too many, it will try to lose them. Other atoms that only need 1 or 2 more electrons are happy to pick them up!

Let's look at a water molecule as an example.

These hydrogen atoms only have 1 electron in a shell that can hold 2, so they each want to get one more



This oxygen atom starts with 6 electrons in a shell that can hold 8, so it wants 2 more. When they share electrons, they are all happy!

How to Make a Bohr Model

1. Write the number of protons and neutrons in the nucleus. You can find this information on the period square
2. Fill in the electrons starting from the innermost orbital and working outward. Remember the maximum number of electrons each orbital shell can hold!